

Research Plan for Doctoral Degree: Untapping The Hidden Potential of Non-Textual Maritime Data Using AI

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1 Introduction

This thesis addresses the challenge of organizing, filtering, searching, and retrieving information from unstructured, non-textual data, including audio, video, images, and 3D models. The research is applied in maritime simulation and virtual commissioning to extract and organize data used for automated testing.

1.1 Research problems and questions

This research addresses the following research problems:

Problem 1: As industries like maritime and automotive generate massive amounts of audio and video data, only a small fraction of this data proves relevant for training artificial intelligence (AI) models. Current methods for storing and querying unstructured data, such as traditional databases, struggle to efficiently manage the volume of video data produced.

Problem 2: Engineering data is rarely available in a machine-readable structured form. It is often represented in formats like figures, drawings, or diagrams, making data extraction and integration challenging. Thus, automatic extraction of information and AI integration in engineering tools is challenging.

This thesis aims to apply AI-driven techniques to organize, filter, and retrieve relevant information from non-textual data and represent it in a machine-readable format. This enables, for example, natural language querying and graph search methods for retrieving information. In this work, we aim to answer the following research questions:

Question 1: What is the state-of-the-art in AI for enhancing the retrieval of unstructured non-textual data in maritime simulation environments?

Question 2: Can complex engineering and maritime unstructured data be converted into machine-readable structures for easier retrieval and integration of information?

1.2 Research aim and scope

The two key areas of the research areas:

From Unstructured to Structured Data Formats: Investigating methods to organize, store, and retrieve information from large-scale unstructured non-textual data such as videos, audio, or figures.

Represent Data in Machine-Understandable Formats: Find an intermediate data structure for representing the information in non-textual unstructured data that is machine-readable and searchable.

Data Integration: Develop methods for encoding dependencies between data sources and integrating them in traditional engineering tools.

The scope is limited to organizing existing unstructured data formats and integrating them into testing and simulation frameworks. This work does not aim to create or propose new data standards or formats to make extraction easier.

1.3 Research methods

We aim to apply AI methods to address the research Problems 1 & 2 and answer Questions 1 & 2. Examples of AI methods for organizing, representing, and retrieving information are listed below:

Natural language processing Natural Language Processing (NLP) is a machine learning technology that enables computers to interpret, manipulate, and understand human language with the help of large language models (LLMs).

Multimodal embeddings Embedding encodes information as vectors embedded into a vector space, allowing the calculation of similarities between objects. Multimodal embeddings encode information of different formats as vectors, allowing us to compare, e.g., images to text paragraphs.

Semantic search Semantic search methods involve searching for meaning, typically using natural language, instead of lexical search, where the search engine looks for exact matches to indexed keywords.

Quantitative and qualitative analyses of secondary data are to be performed. Data is mainly obtained from the maritime industry and training activities at the Aboa Mare Training Academy through ongoing research projects.

1.4 State-of-the-art review

Image and video data analytics are essential in maritime systems, providing situational awareness on vessels and at remote monitoring centres [1]. However, the rapid increase in collected unstructured data presents significant challenges: the constant influx of data overwhelms storage systems,

creating scalability issues and making it difficult to extract relevant data for research, such as training machine learning models or generating simulation scenarios.

In image and video processing, models like convolutional neural networks (CNNs) [2] for image classification and object detection, as well as advanced architectures like Vision Transformers (ViT) [3, 4], are widely used, especially for tasks that require spatial relationship understanding [5]. Processing video data requires considerable computational resources and often benefits from GPU acceleration.

Audio data, on the other hand, presents its own set of challenges. Speech-to-text conversion for spoken content and specialized techniques for non-speech sounds are essential. Deep learning models like OpenAI Whisper [6], wav2vec [7], and HuBERT [8] have shown promising results in this domain including [9, 10]. Moreover, large language models (LLMs) such as GPT-4 [11] and its successors have demonstrated exceptional capabilities in understanding and generating human-like text.

As recent maritime and automotive industries increasingly deal with various forms of unstructured data, the importance of *multi-modal AI models* that can simultaneously process and interpret different data types is growing. Models like CLIP (Contrastive Language-Image Pre-training) [12] showcase the potential of combining text and image understanding for more comprehensive analysis.

When retrieving specific information from unstructured data, converting it into an intermediate, machine-readable, and searchable format is beneficial. Examples of such task-specific standardized formats include the DEXPI open standard [13] for industrial design data, like P&ID diagrams, and the OpenUSD [14] for representing 3D scenes.

2 Research environment

The research will be conducted in an international, collaborative research environment consisting of a consortium of research organizations and industries with expertise in maritime technology and simulation. The research will be funded by the Co-Innovation project Virtual Sea Trial and executed under a contract with Novia University of Applied Sciences. Part of this work will also be done in the joint COSWIM project between Novia UAS and the RISE Research Institutes of Sweden. Selected collaborators and partners are listed below.

Novia Intelligent Systems Institute promotes collaboration with Finnish businesses by leveraging the collective expertise in AI, machine learning, and information management. Its team includes project experts, researchers from Novia UAS, and a group of teachers.

Virtual Sea Trial is a Business Finland-funded Co-Innovation research project for the period 1.1.2024–31.12.2026, consisting of four university partners and eight industry partners.

COSWIM is a project coordinated by RISE Research Institutes of Sweden and focuses on Co-simulation for maritime remote operations.

DDSM is a Business Finland-funded project coordinated by ÅAU focusing on data-driven sustainability management. This research can also utilize the data management methodology developed in DDSM.

The Finnish Software Engineering Doctoral Research Network aims to improve doctoral education in software engineering in Finland, aligning it with real-world industry and societal needs, and promoting higher-level academic-industry collaboration. Åbo Akademi University (ÅAU) coordinates the network.

Nordic Cooperation on Maritime Autonomous Surface Ships (NorMASS) is an academic collaboration that explores new horizons in teaching, research, and simulation technology dedicated to Maritime Autonomous Surface Ships (MASS).

3 Implementation and Publication plan

The research will be conducted in the Virtual Sea Trial project for the period 1.1.2024–31.12.2026. The presented publication plan is in chronological order. The goal is to publish **Paper I** and **Paper II** in the first year of the PhD studies, **Paper III** in the second, and **Paper IV** in the third.

Paper I: *Maritime Automatic Speech Recognition and Key-Phrase Extraction.* This paper focuses on developing an automatic speech recognition (ASR) framework for maritime VHF radio communications. The objective is to transcribe audio and recognize selected key phrases in real time.

Paper II: *Semantic Search for Maritime Unstructured Data.* This paper will explore the uses of natural-language queries to address the challenge of retrieving relevant data from unstructured sources, including PDF documents, audio data, and video sequences from large databases.

Paper III: *Graph Representations of Maritime Systems.* This paper explores using relational graph representations to model the interaction between ship onboard systems. The objective is to enable efficient search and retrieval of information from subsystems and components to support automated test generation and simulation.

Paper IV: *Extracting Information from Engineering Graphs and Diagrams.* This paper addresses the challenge of extracting specific information, such as parameter values or operating conditions, from engineering charts and P&ID diagrams.

3.1 Planned research exchange

Novia UAS will provide the possibility for a 3–6 month research exchange in the second year (2026) of the PhD studies.

4 Discussion

The maritime industry and technology providers are expected to benefit from the results through increased operational efficiency and cost-effectiveness. Results happen through publication in (prefer-

ably) open-access journals and conference series. Dissemination of results happens through the Virtual Sea Trial project.

References

- [1] S. Thombre, Z. Zhao, H. Ramm-Schmidt, J. M. V. García, T. Malkamäki, S. Nikolskiy, T. Hammarberg, H. Nuortie, M. Z. H. Bhuiyan, S. Särkkä, *et al.*, “Sensors and AI techniques for situational awareness in autonomous ships: A review,” *IEEE transactions on intelligent transportation systems*, vol. 23, no. 1, pp. 64–83, 2020.
- [2] N. Ketkar, J. Moolayil, N. Ketkar, and J. Moolayil, “Convolutional neural networks,” *Deep learning with Python: learn best practices of deep learning models with PyTorch*, pp. 197–242, 2021.
- [3] X. Zhai, A. Kolesnikov, N. Houlsby, and L. Beyer, “Scaling vision transformers,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 12104–12113, June 2022.
- [4] L. Yuan, Y. Chen, T. Wang, W. Yu, Y. Shi, Z.-H. Jiang, F. E. Tay, J. Feng, and S. Yan, “Tokens-to-token vit: Training vision transformers from scratch on imagenet,” in *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 558–567, 2021.
- [5] A. Thippur, C. Burbridge, L. Kunze, M. Alberti, J. Folkesson, P. Jensfelt, and N. Hawes, “A comparison of qualitative and metric spatial relation models for scene understanding,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 29, 2015.
- [6] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Robust speech recognition via large-scale weak supervision,” in *International Conference on Machine Learning*, pp. 28492–28518, PMLR, 2023.
- [7] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, “wav2vec 2.0: A framework for self-supervised learning of speech representations,” *Advances in neural information processing systems*, vol. 33, pp. 12449–12460, 2020.
- [8] W.-N. Hsu, B. Bolte, Y.-H. H. Tsai, K. Lakhotia, R. Salakhutdinov, and A. Mohamed, “Hubert: Self-supervised speech representation learning by masked prediction of hidden units,” *IEEE/ACM transactions on audio, speech, and language processing*, vol. 29, pp. 3451–3460, 2021.
- [9] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, “Roberta: A robustly optimized BERT pretraining approach,” *arXiv preprint arXiv:1907.11692*, 2019.
- [10] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “Bert: Pre-training of deep bidirectional transformers for language understanding,” *arXiv preprint arXiv:1810.04805*, 2018.
- [11] “GPT-4 technical report,” 2024.

- [12] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Aspell, P. Mishkin, J. Clark, *et al.*, “Learning transferable visual models from natural language supervision,” in *International conference on machine learning*, pp. 8748–8763, PMLR, 2021.
- [13] J. Oeing, W. Welscher, N. Krink, L. Jansen, F. Henke, and N. Kockmann, “Using artificial intelligence to support the drawing of piping and instrumentation diagrams using dexpi standard,” *Digital Chemical Engineering*, vol. 4, p. 100038, 2022.
- [14] “Universal scene description.” <https://openusd.org/release/index.html>. Accessed: 2024-10-29.